

RNA-Seq data analysis Part2

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Differential gene expression

- **Goal:** Detection of statistically significant differences between conditions or time points
- Normalization step
- Computation of the fold-change

$$FC = \frac{expression_{treatment}}{expression_{control}}$$

- Statistical testing
 - assessing the p-value / corrected p-value for multiple testing

Common approach for statistical testing

- test the null hypothesis that $\log_2(\text{FC})$ between treatment and control for a gene's expression is exactly zero, i.e., that the gene is not at all affected by the treatment.
- output: list of genes ranked by p-value (corrected for multiple testing)
- issues:
 - genes with low counts
 - varying sample sizes

DESeq2 model

- uses a generalised linear model of the form

$$K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$$

$$\mu_{ij} = s_j q_{ij}$$

$$\log_2(q_{ij}) = x_j \beta_i$$

where counts K_{ij} for gene i in sample j are modelled using a negative binomial distribution with fitted mean μ_{ij} and a gene-specific dispersion parameter α_i .

s_j = sample specific size factor (normalisation)

q_{ij} = parameter proportional to the expected true concentration of fragments for sample j

Dispersion

- represents the extent to which a distribution is stretched or squeezed
- measures of statistical dispersion (e.g. variance, standard deviation)
- Assumption: genes of similar average expression strength have similar dispersion
- dispersion parameter α_i defines the relationship between the variance of the observed counts and its mean value: (within group variability)

$$\text{Var}(K_{ij}) = \mu_{ij} + \alpha_i \mu_{ij}^2$$

- DESeq2 detects and corrects dispersion estimates that are too low through modeling of the dependence of the dispersion on the average expression strength over all samples.

DESeq2 steps for DEG

- estimation of the size factors s_j (see the normalisation)
- estimation of dispersion α_i
- negative binomial generalised model fitting for each gene (β_i)
- test with a Wald test whether each model coefficient differs significantly from zero.

Wald test

- compares the beta estimate β_i (the shrunken estimate of $\log_2(\text{FC})$ divided by its standard error to a standard normal distribution.
- two-tailed test
- the reported Wald test p-values are adjusted for multiple testing using the Benjamini-Hochberg correction.